

Artificial Intelligence Lab

Iterative Deepening Search (IDS)

Problem Statement

Write a Python program to implement **Iterative Deepening Search (IDS)** for a graph. The graph, start node, and goal node should be taken as input from the user.

Your program should repeatedly perform a **Depth-Limited Search** with depth limits $0, 1, 2, \dots$ until the goal is found.

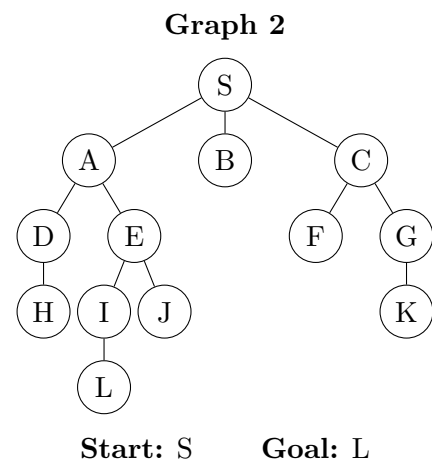
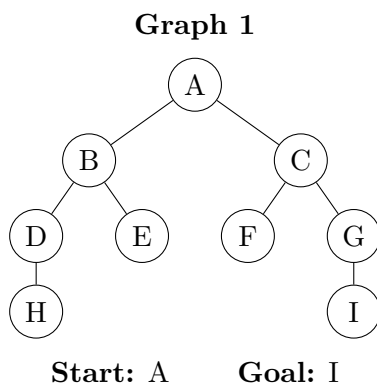
For each depth limit, display:

- the current depth limit,
- the nodes visited at that limit,
- and, when found, the path from the start node to the goal node.

Suggested functions:

- `depth_limited_search(...)`
- `iterative_deepening_search(...)`

Graphs to Try



Note: Assume neighbours are explored in the order in which they are entered. Test your program on both of the above graphs.